Unit 2 - Entity Relationship Model

Database Management (DI03016041) - GTU Diploma IT Sem 3

20% weightage | 9 hours

How to use these notes: every topic is written in a descriptive, copy-ready way. **3 marks** = definition + 2-3 points. **4 marks** = add example or table. **7 marks** = write everything with diagram / example. Lines marked ★ show where the topic was asked in a GTU paper (Winter 2025 / Summer 2026). SQL examples use Oracle syntax.

GTU questions already asked from this unit (do these first)

- W2025 Q2(a) [Relationship, Weak entity set, Primary key] + Q2(a)OR [Entity, Recursive relationship set, Super key] + S2026 Q2(a)OR [Entity, Attributes, Relationship]
- Asked in BOTH papers (7M each): W2025 Q2(c) [attribute and its types] + S2026 Q2(c)OR [define attribute, types with example]
- Key terms asked in W2025 Q2(a) [Primary key] + Q2(a)OR [Super key]
- Asked in BOTH papers: W2025 Q2(b)OR [4M: mapping cardinality with example] + S2026 Q2(a) [3M: with diagram]
- Asked in BOTH papers (7M each): W2025 Q2(c)OR + S2026 Q2(c) [ER diagram and symbols with example]
- Weak entity set asked in W2025 Q2(a) [3M]
- Asked in BOTH papers: W2025 Q2(b) [4M: generalization with example] + S2026 Q2(b)OR [4M: describe generalization]
- Asked in S2026 Q2(b) [4M: specialization with example]

2.1 Basic concepts of E-R: entity, attributes, relationship, participation, recursive, degree

★ W2025 Q2(a) [Relationship, Weak entity set, Primary key] + Q2(a)OR [Entity, Recursive relationship set, Super key] + S2026 Q2(a)OR [Entity, Attributes, Relationship]

The **Entity-Relationship (E-R) model** (Peter Chen, 1976) is a **high-level conceptual data model** that describes the database as a collection of **entities** and the **relationships** among them. It is used to design the database before creating tables.

- **Entity:** a real-world **object or thing that can be distinguished** from others and about which data is stored. *Example: student, teacher, course, car.* An **entity set** is a collection of similar entities (all students).
- **Relationship:** an **association between two or more entities**. *Example: Student ENROLLS in Course; Teacher TEACHES Subject.* A relationship set is a set of similar relationships.
- **Attribute:** a **property / characteristic** that describes an entity or a relationship. *Example: Student has Roll_no, Name, City.*

Participation (2.1.3.1)

- **Total participation** (shown by a **double line**): **every** entity of the set must take part in the relationship. *Every Loan must belong to some Customer.*

- **Partial participation** (single line): **some** entities may not take part. *Not every Customer has a Loan.*

Recursive relationship (2.1.3.2)

A relationship in which the **same entity set participates more than once in different roles** (also called unary relationship). *Example: EMPLOYEE '*manages*' EMPLOYEE - one employee is Manager, the other is Worker.*

Degree of a relationship set (2.1.3.3)

The **number of entity sets that participate** in the relationship.

Degree	Name	Example
1	Unary (recursive)	Employee manages Employee
2	Binary (most common)	Student enrolls in Course
3	Ternary	Doctor - Patient - Medicine (prescribes)

Term	Definition (3-mark style)
Entity	A real-world object or thing, distinguishable from others, about which data is stored (e.g. Student, Book).
Attribute	A property that describes an entity (e.g. Name, Roll_no of Student).
Relationship	An association among two or more entities (e.g. Student *enrolls* in Course).
Recursive relationship set	A relationship in which the same entity set takes part more than once in different roles (e.g. Employee manages Employee).
Weak entity set	Entity set without its own primary key; depends on an owner entity (e.g. Dependent of Employee).
Primary key	The candidate key chosen to uniquely identify each row; cannot be NULL or duplicate.
Super key	A set of one or more attributes that uniquely identifies a row; may contain extra attributes (e.g. {Roll_no, Name}).

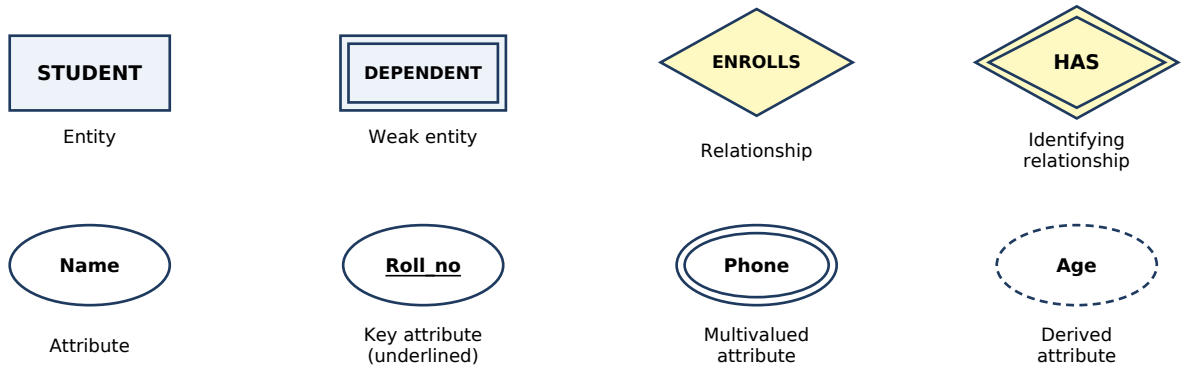
2.1.2 Attributes and their types

★ Asked in BOTH papers (7M each): W2025 Q2(c) [attribute and its types] + S2026 Q2(c)OR [define attribute, types with example]

An **attribute** is a **property or characteristic of an entity (or relationship)**. Each attribute has a **domain** (set of allowed values). Example: for entity STUDENT the attributes are Roll_no, Name, Address, Phone, DOB.

Types of attributes

Type	Meaning	Example	ER symbol
Simple (atomic)	Cannot be divided further	Roll_no, Age, Gender	Single ellipse
Composite	Can be divided into smaller sub-parts	Name → First_name, Middle_name, Last_name; Address → Street, City, Pin	Ellipse with child ellipses
Single-valued	Has only **one value** for an entity	Roll_no, DOB	Single ellipse
Multi-valued	Can have **many values** for one entity	Phone_numbers, Email_ids, Skills	**Double ellipse**
Stored	Value is directly stored in the database	Date_of_birth	Single ellipse
Derived	Value is **calculated from other attribute**; need not be stored	Age (from DOB); Total_marks	**Dashed ellipse**
Key attribute	Uniquely identifies each entity	Roll_no for STUDENT	**Underlined** ellipse
Null attribute	Value unknown or not applicable (NULL)	Middle_name, Passport_no	-
Complex	Combination of composite and multi-valued	Address_history	-



Straight line = links attributes to entities and entities to relationships; 1, M, N near the line show cardinality

Keys: super, candidate, primary, foreign

★ Key terms asked in W2025 Q2(a) [Primary key] + Q2(a)OR [Super key]

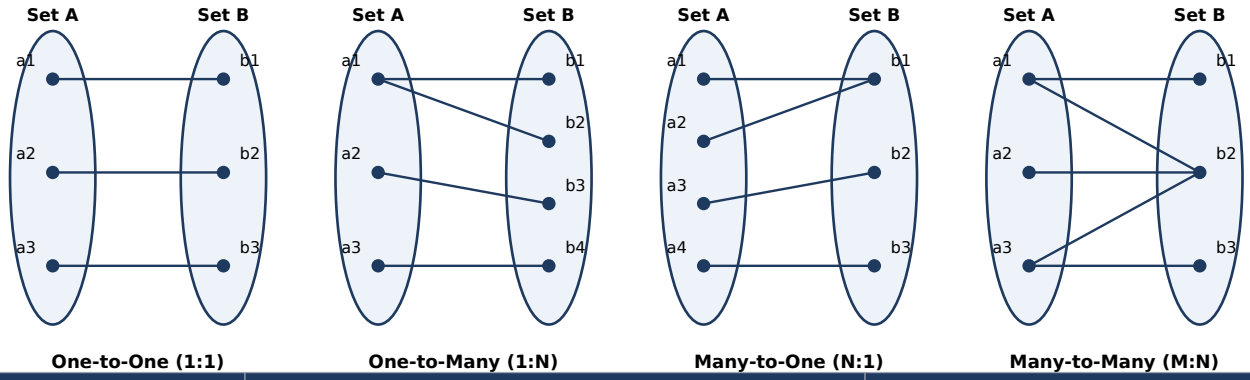
A **key** is an attribute (or a set of attributes) that **uniquely identifies** a row in a table.

Key	Definition	Example (STUDENT: Roll_no, Enrol_no, Name, Phone)
Super key	Any attribute or set of attributes that uniquely identifies a row (may contain extra attributes)	{Roll_no}, {Roll_no, Name}, {Enrol_no, Phone}
Candidate key	A **minimal super key** - no attribute can be removed	{Roll_no}, {Enrol_no}
Primary key	The candidate key **chosen** by the designer to identify rows; cannot be NULL or duplicate; only one per table	Roll_no
Alternate key	Candidate keys not chosen as primary key	Enrol_no
Composite key	Primary key made of two or more attributes	(StudentID, CourseID) in ENROLLMENT
Foreign key	Attribute in one table that refers to the primary key of another table; links tables	Dept_id in STUDENT referring to DEPARTMENT(Dept_id)

2.2 Mapping cardinality

★ Asked in BOTH papers: W2025 Q2(b)OR [4M: mapping cardinality with example] + S2026 Q2(a) [3M: with diagram]

Mapping cardinality (cardinality ratio) tells **how many entities of one set can be associated with how many entities of another set** through a relationship set. For a binary relationship between entity sets A and B there are four types:



One-to-One (1:1) | One-to-Many (1:N) | Many-to-One (N:1) | Many-to-Many (M:N)

Type	Meaning	Example
One-to-One (1:1)	An entity in A is related to **at most one** entity in B and vice versa	Person - Passport; College - Principal
One-to-Many (1:N)	An entity in A can be related to **many** entities in B; an entity in B to only one in A	Department - Students (one department has many students)
Many-to-One (N:1)	Many entities in A relate to **one** entity in B	Employees - Department
Many-to-Many (M:N)	Many in A relate to many in B	Student - Course; Author - Book

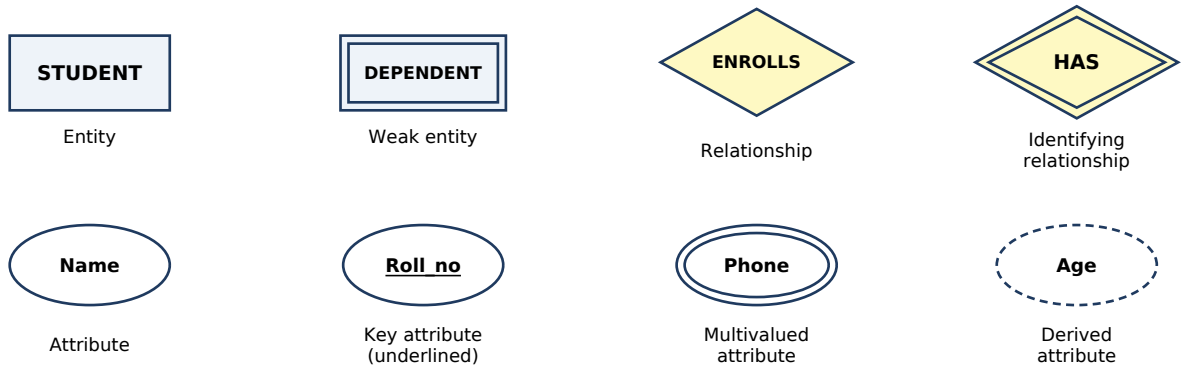
2.3 ER diagrams and symbols

★ Asked in BOTH papers (7M each): W2025 Q2(c)OR + S2026 Q2(c) [ER diagram and symbols with example]

An **E-R diagram** is a **graphical (pictorial) representation** of the entities, their attributes and the relationships among them in a database. It is the blueprint used to design the database.

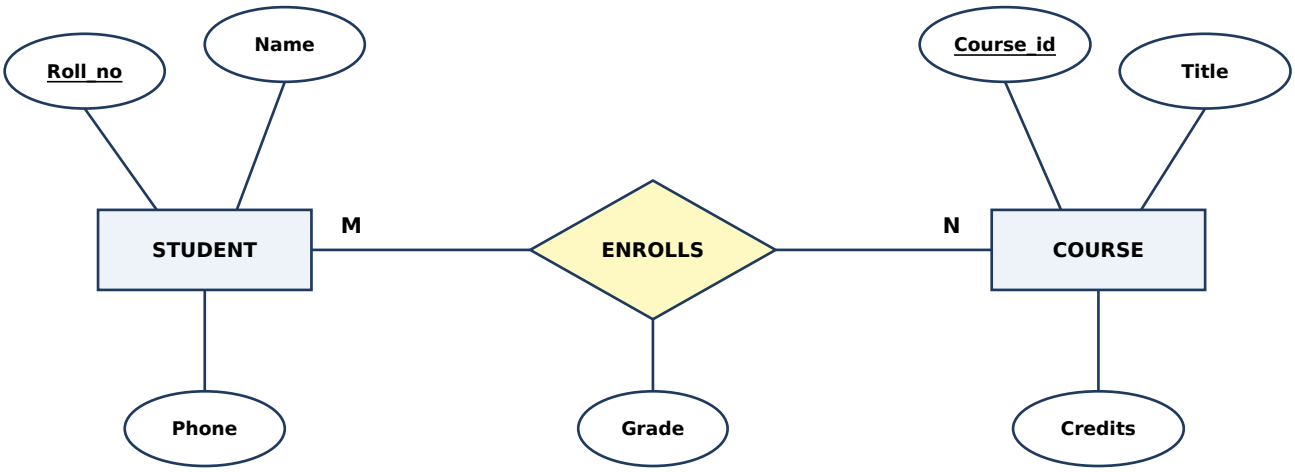
Symbols used in an E-R diagram

Symbol	Name	Meaning
Rectangle	Entity set	A real-world object (STUDENT, COURSE)
Double rectangle	Weak entity set	Entity that has no key of its own and depends on a strong entity (DEPENDENT)
Ellipse (oval)	Attribute	Property of an entity or relationship
Underlined ellipse	Key attribute	Uniquely identifies entities (Roll_no)
Double ellipse	Multi-valued attribute	Can have many values (Phone_no)
Dashed ellipse	Derived attribute	Calculated from other attributes (Age)
Diamond	Relationship set	Association between entities (ENROLLS)
Double diamond	Identifying relationship	Relationship between a weak entity and its owner entity
Line	Link	Connects attributes to entities and entities to relationships
Double line	Total participation	Every entity must participate in the relationship
1, M, N on lines	Cardinality	Shows 1:1, 1:N, M:N



Straight line = links attributes to entities and entities to relationships; 1, M, N near the line show cardinality

Example: Student - Course



Many students enrol in many courses (M : N relationship)

Reading the diagram: entity STUDENT (key Roll_no, attributes Name, Phone) is connected to entity COURSE (key Course_id, attributes Title, Credits) through the relationship ENROLLS. The cardinality is M:N and the relationship has its own attribute Grade.

2.4 Weak entity sets

★ Weak entity set asked in W2025 Q2(a) [3M]

A **weak entity set** is an entity set that **does not have a primary key of its own**. Its existence depends on another entity called the **owner (strong / identifying) entity**.

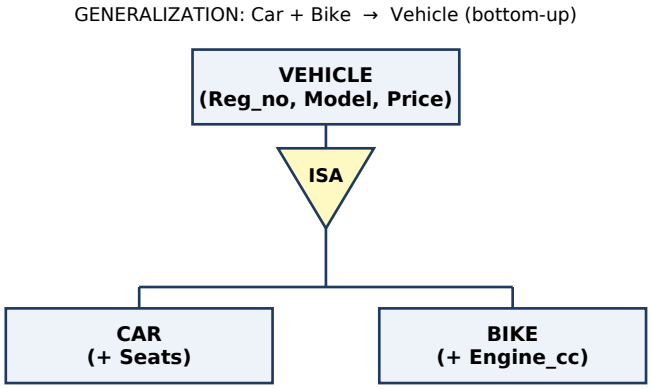
- It has a **partial key (discriminator)** - shown with a **dashed underline**; the primary key of a weak entity = primary key of owner + partial key.
- Drawn as a **double rectangle**; its relationship with the owner is a **double diamond** (identifying relationship) with total participation.
- **Example:** DEPENDENT (name, age, relation) of an EMPLOYEE - a dependent cannot exist without the employee; two employees may have dependents with the same name, so Name alone cannot identify a dependent.

Strong entity = has its own primary key (EMPLOYEE).

2.5 Enhanced ER: generalization

★ Asked in BOTH papers: W2025 Q2(b) [4M: generalization with example] + S2026 Q2(b)OR [4M: describe generalization]

Generalization is a **bottom-up** process in which two or more lower-level entity sets that share common attributes are **combined into a higher-level (super class) entity set**.

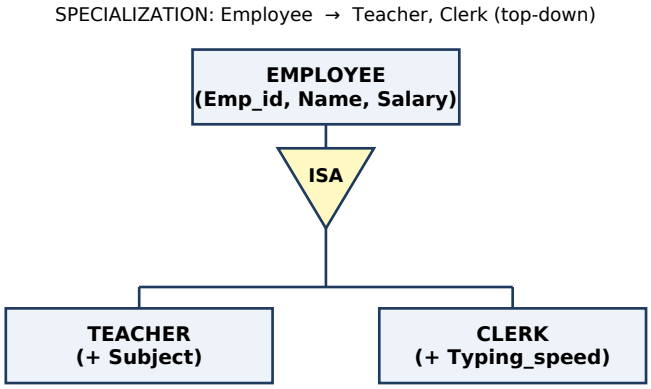


- Common attributes (Reg_no, Model, Price) are moved up to the super class VEHICLE; special attributes stay with CAR and BIKE.
- The relationship is shown by an **ISA** triangle (CAR *is a* VEHICLE).
- **Purpose:** removes duplication of attributes and shows common features.

2.5 Enhanced ER: specialization

★ Asked in S2026 Q2(b) [4M: specialization with example]

Specialization is a **top-down** process in which a **higher-level entity set is divided into lower-level (sub class) entity sets** based on some distinguishing characteristic.



- Sub classes **inherit all attributes** of the super class and add their own specific attributes.
- **Example:** EMPLOYEE is specialised into TEACHER (extra attribute Subject) and CLERK (extra attribute Typing_speed).
- **Generalization vs specialization:** opposite processes - generalization is bottom-up (combine), specialization is top-down (split).

2.5 Subclass, super class and aggregation

- **Super class and sub class:** super class = general higher-level entity (EMPLOYEE); sub class = specialised lower-level entity (TEACHER). Sub class inherits attributes and relationships of its super class.
- **Aggregation:** an abstraction in which a **relationship set is treated as a higher-level entity** so that it can take part in another relationship. *Example: the relationship 'Employee WORKS_ON Project' is aggregated, and then a MANAGER 'monitors' this whole work-on relationship.*

2.6 Converting ER diagrams to database (tables)

Rules for converting an ER diagram to tables (relational schema):

ER element	Conversion rule	Example
Strong entity	Make a table; attributes become columns; key attribute becomes primary key	STUDENT(Roll_no, Name, City)
Composite attribute	Take only its simple components as columns	Name → First_name, Last_name
Multi-valued attribute	Make a **separate table** with the entity's key + the attribute	STUDENT_PHONE(Roll_no, Phone)
Derived attribute	Not stored	Age is not a column
1:1 relationship	Put primary key of one side as foreign key in the other	-
1:N relationship	Put the primary key of the '1' side as foreign key in the 'N' side table	STUDENT(Roll_no, ..., Dept_id FK)
M:N relationship	Make a **new table** with primary keys of both entities (as composite key) + relationship attributes	ENROLLS(Roll_no, Course_id, Grade)
Weak entity	Table with owner's primary key + partial key as composite primary key	DEPENDENT(Emp_id, Dep_name, Age)
Generalization / specialization	Either one table per sub class (with super-class key) or one combined table	TEACHER(Emp_id, Subject)

Underlined = primary key. FK = foreign key.

Last-minute revision sheet - Unit 2

- Entity = object (rectangle); attribute = property (ellipse); relationship = association (diamond).
- Attribute types: simple, composite, single-valued, multi-valued (double ellipse), derived (dashed), key (underlined), stored, null.
- Mapping cardinality: 1:1, 1:N, N:1, M:N.
- Weak entity: no own key; double rectangle; depends on owner via identifying relationship (double diamond).
- Generalization = bottom-up (combine); specialization = top-down (split); ISA triangle.
- ER to table: entity → table; M:N relationship → new table; multi-valued attribute → new table; 1:N → FK on N side.